



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | drum               |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.12.1 |
| Date            | 02/21/2025         |

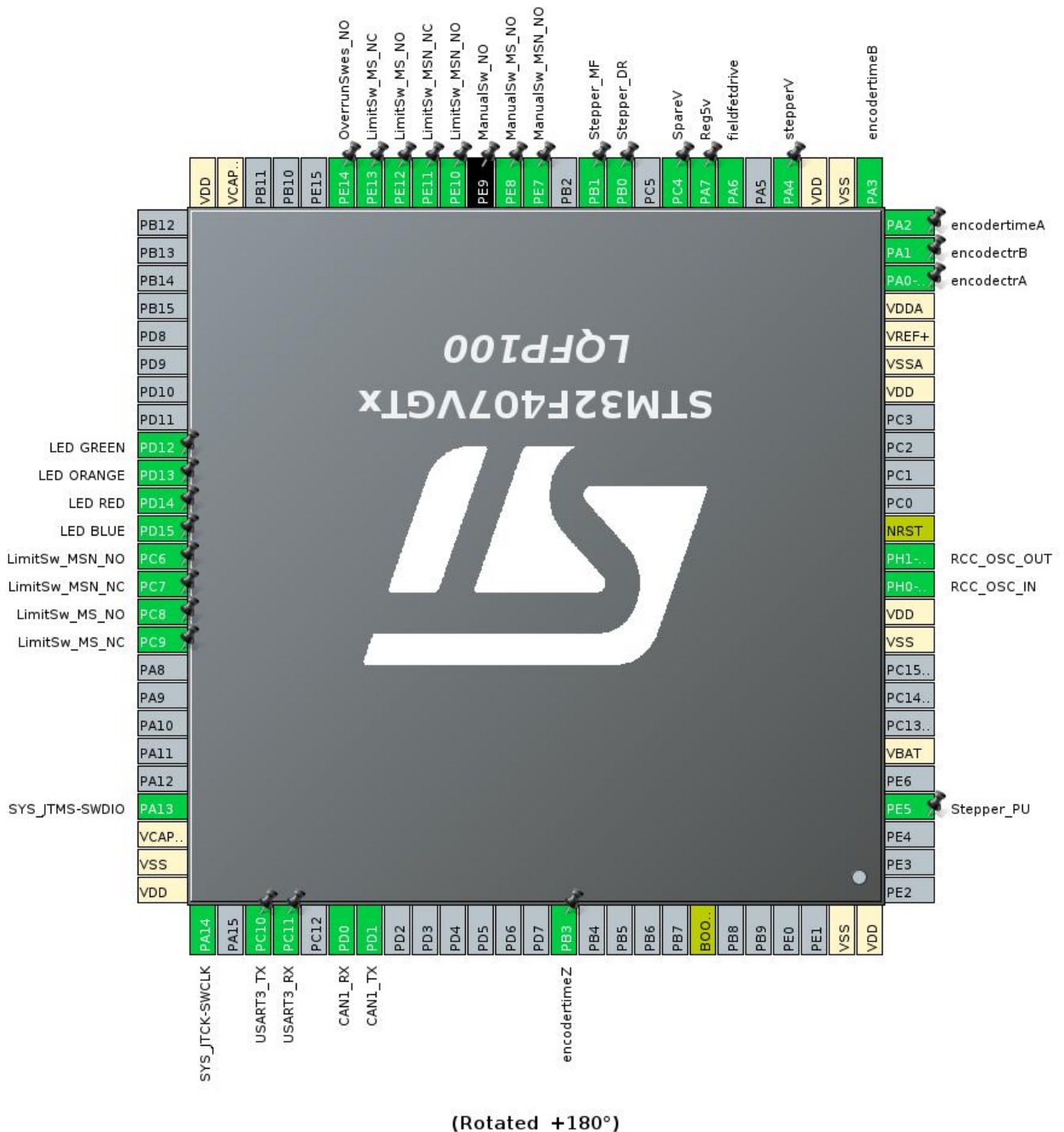
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F407/417 |
| MCU name       | STM32F407VGTx |
| MCU Package    | LQFP100       |
| MCU Pin number | 100           |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



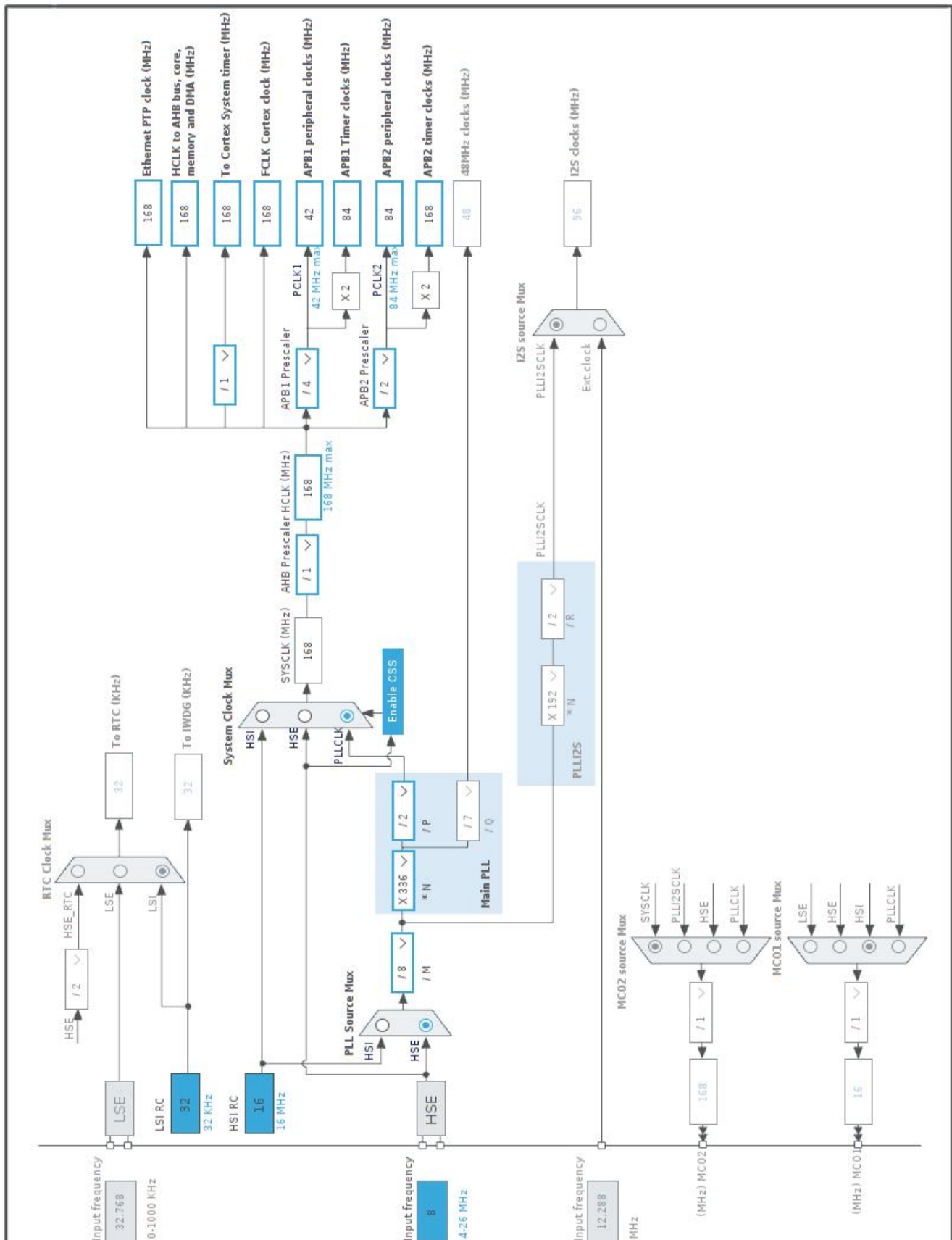
### 3. Pins Configuration

| Pin Number<br>LQFP100 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label           |
|-----------------------|---------------------------------------|----------|--------------------------|-----------------|
| 4                     | PE5                                   | I/O      | TIM9_CH1                 | Stepper_PU      |
| 6                     | VBAT                                  | Power    |                          |                 |
| 10                    | VSS                                   | Power    |                          |                 |
| 11                    | VDD                                   | Power    |                          |                 |
| 12                    | PH0-OSC_IN                            | I/O      | RCC_OSC_IN               |                 |
| 13                    | PH1-OSC_OUT                           | I/O      | RCC_OSC_OUT              |                 |
| 14                    | NRST                                  | Reset    |                          |                 |
| 19                    | VDD                                   | Power    |                          |                 |
| 20                    | VSSA                                  | Power    |                          |                 |
| 21                    | VREF+                                 | Power    |                          |                 |
| 22                    | VDDA                                  | Power    |                          |                 |
| 23                    | PA0-WKUP                              | I/O      | TIM5_CH1                 | encodectrA      |
| 24                    | PA1                                   | I/O      | TIM5_CH2                 | encodectrB      |
| 25                    | PA2                                   | I/O      | TIM2_CH3                 | encodertimeA    |
| 26                    | PA3                                   | I/O      | TIM2_CH4                 | encodertimeB    |
| 27                    | VSS                                   | Power    |                          |                 |
| 28                    | VDD                                   | Power    |                          |                 |
| 29                    | PA4                                   | I/O      | ADC1_IN4                 | stepperV        |
| 31                    | PA6                                   | I/O      | TIM13_CH1                | fieldfetdrive   |
| 32                    | PA7                                   | I/O      | ADC1_IN7                 | Reg5v           |
| 33                    | PC4                                   | I/O      | ADC1_IN14                | SpareV          |
| 35                    | PB0 *                                 | I/O      | GPIO_Output              | Stepper_DR      |
| 36                    | PB1 *                                 | I/O      | GPIO_Output              | Stepper_MF      |
| 38                    | PE7 *                                 | I/O      | GPIO_Input               | ManualSw_MSN_NO |
| 39                    | PE8 *                                 | I/O      | GPIO_Input               | ManualSw_MS_NO  |
| 40                    | PE9 *                                 | I/O      | GPIO_Input               | ManualSw_NO     |
| 41                    | PE10                                  | I/O      | GPIO_EXTI10              | LimitSw_MSN_NO  |
| 42                    | PE11                                  | I/O      | GPIO_EXTI11              | LimitSw_MSN_NC  |
| 43                    | PE12                                  | I/O      | GPIO_EXTI12              | LimitSw_MS_NO   |
| 44                    | PE13                                  | I/O      | GPIO_EXTI13              | LimitSw_MS_NC   |
| 45                    | PE14 *                                | I/O      | GPIO_Input               | OverrunSwes_NO  |
| 49                    | VCAP_1                                | Power    |                          |                 |
| 50                    | VDD                                   | Power    |                          |                 |
| 59                    | PD12 *                                | I/O      | GPIO_Output              | LED GREEN       |
| 60                    | PD13 *                                | I/O      | GPIO_Output              | LED ORANGE      |
| 61                    | PD14 *                                | I/O      | GPIO_Output              | LED RED         |

| Pin Number<br>LQFP100 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label          |
|-----------------------|---------------------------------------|----------|--------------------------|----------------|
| 62                    | PD15 *                                | I/O      | GPIO_Output              | LED BLUE       |
| 63                    | PC6                                   | I/O      | TIM3_CH1                 | LimitSw_MSN_NO |
| 64                    | PC7                                   | I/O      | TIM3_CH2                 | LimitSw_MSN_NC |
| 65                    | PC8                                   | I/O      | TIM3_CH3                 | LimitSw_MS_NO  |
| 66                    | PC9                                   | I/O      | TIM3_CH4                 | LimitSw_MS_NC  |
| 72                    | PA13                                  | I/O      | SYS_JTMS-SWDIO           |                |
| 73                    | VCAP_2                                | Power    |                          |                |
| 74                    | VSS                                   | Power    |                          |                |
| 75                    | VDD                                   | Power    |                          |                |
| 76                    | PA14                                  | I/O      | SYS_JTCK-SWCLK           |                |
| 78                    | PC10                                  | I/O      | USART3_TX                |                |
| 79                    | PC11                                  | I/O      | USART3_RX                |                |
| 81                    | PD0                                   | I/O      | CAN1_RX                  |                |
| 82                    | PD1                                   | I/O      | CAN1_TX                  |                |
| 89                    | PB3                                   | I/O      | TIM2_CH2                 | encodertimeZ   |
| 94                    | BOOT0                                 | Boot     |                          |                |
| 99                    | VSS                                   | Power    |                          |                |
| 100                   | VDD                                   | Power    |                          |                |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F407/417 |
| MCU       | STM32F407VGTx |
| Datasheet | DS8626 Rev8   |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.3 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

#### 1.4. Sequence

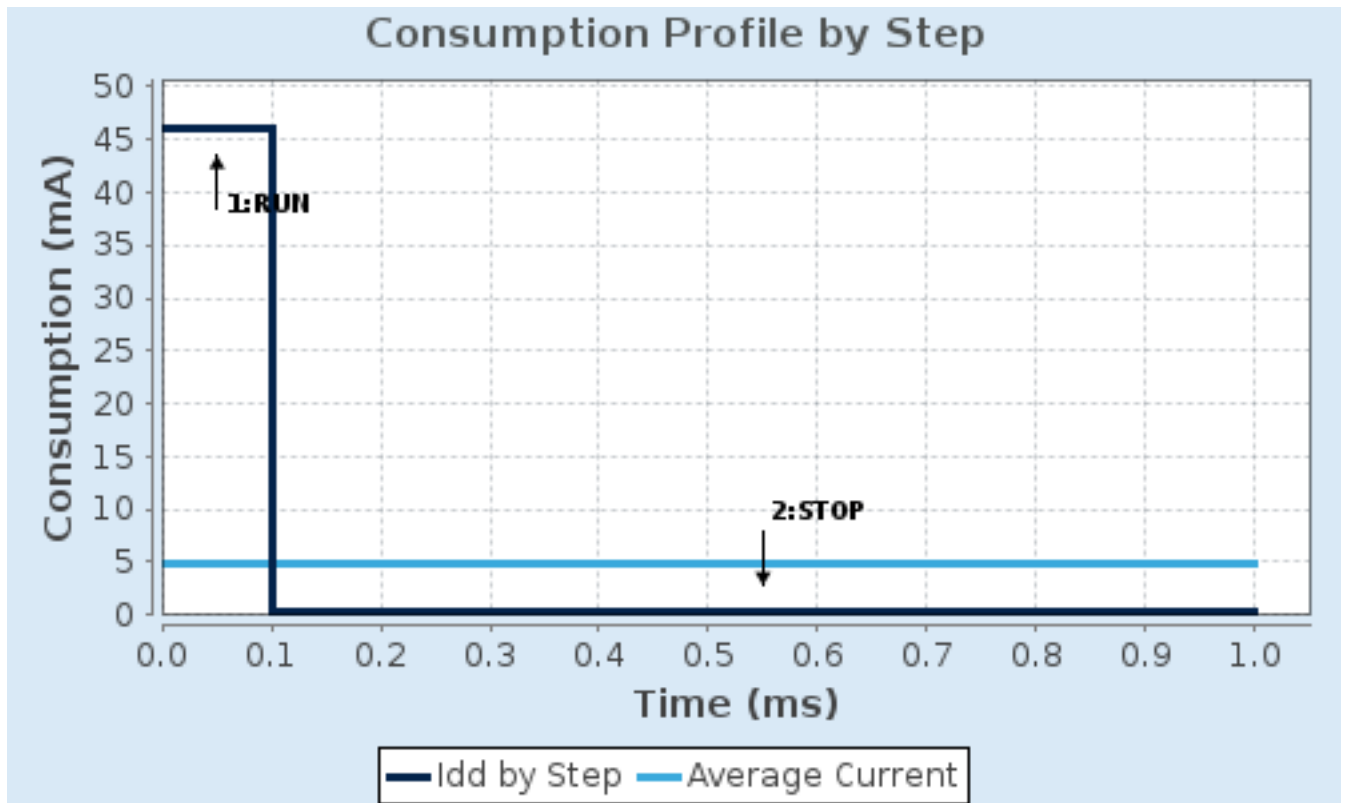
|                               |             |                           |
|-------------------------------|-------------|---------------------------|
| <b>Step</b>                   | Step1       | Step2                     |
| <b>Mode</b>                   | RUN         | STOP                      |
| <b>Vdd</b>                    | 3.3         | 3.3                       |
| <b>Voltage Source</b>         | Battery     | Battery                   |
| <b>Range</b>                  | Scale1-High | No Scale                  |
| <b>Fetch Type</b>             | FLASH       | n/a                       |
| <b>CPU Frequency</b>          | 168 MHz     | 0 Hz                      |
| <b>Clock Configuration</b>    | HSE PLL     | Regulator_LP Flash-PwrDwn |
| <b>Clock Source Frequency</b> | 4 MHz       | 0 Hz                      |
| <b>Peripherals</b>            |             |                           |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA                      |
| <b>Average Current</b>        | 46 mA       | 280 $\mu$ A               |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms                    |
| <b>DMIPS</b>                  | 210.0       | 0.0                       |
| <b>Ta Max</b>                 | 98.47       | 104.96                    |
| <b>Category</b>               | In DS Table | In DS Table               |

#### 1.5. Results

|               |                  |                 |             |
|---------------|------------------|-----------------|-------------|
| Sequence Time | 1 ms             | Average Current | 4.85 mA     |
| Battery Life  | 29 days, 4 hours | Average DMIPS   | 210.0 DMIPS |

#### 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                               |
|-----------------------------------|-------------------------------------|
| Project Name                      | drum                                |
| Project Folder                    | /media/deh/fc65a06a-0110-42ac-85a6- |
| Toolchain / IDE                   | Makefile                            |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.28.1             |
| Application Structure             | Basic                               |
| Generate Under Root               | No                                  |
| Do not generate the main()        | No                                  |
| Minimum Heap Size                 | 0x200                               |
| Minimum Stack Size                | 0x400                               |

### 2.2. Code Generation Settings

| Name                                                            | Value                                           |
|-----------------------------------------------------------------|-------------------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy all used libraries into the project folder |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                              |
| Backup previously generated files when re-generating            | No                                              |
| Keep User Code when re-generating                               | Yes                                             |
| Delete previously generated files when not re-generated         | Yes                                             |
| Set all free pins as analog (to optimize the power consumption) | No                                              |
| Enable Full Assert                                              | No                                              |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | MX_GPIO_Init        | GPIO                     |
| 2    | MX_DMA_Init         | DMA                      |
| 3    | SystemClock_Config  | RCC                      |
| 4    | MX_CAN1_Init        | CAN1                     |
| 5    | MX_USART3_UART_Init | USART3                   |
| 6    | MX_TIM2_Init        | TIM2                     |
| 7    | MX_TIM5_Init        | TIM5                     |
| 8    | MX_TIM9_Init        | TIM9                     |
| 9    | MX_TIM13_Init       | TIM13                    |
| 10   | MX_TIM3_Init        | TIM3                     |
| 11   | MX_ADC1_Init        | ADC1                     |

| Rank | Function Name | Peripheral Instance Name |
|------|---------------|--------------------------|
| 12   | MX_TIM4_Init  | TIM4                     |

## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

mode: IN4

mode: IN7

mode: IN14

mode: Temperature Sensor Channel

mode: Vrefint Channel

#### 3.1.1. Parameter Settings:

##### ADCs\_Common\_Settings:

Mode Independent mode

##### ADC\_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution 12 bits (15 ADC Clock cycles)

Data Alignment Right alignment

Scan Conversion Mode Enabled

Continuous Conversion Mode **Enabled \***

Discontinuous Conversion Mode Disabled

DMA Continuous Requests **Enabled \***

End Of Conversion Selection EOC flag at the end of single channel conversion

##### ADC\_Regular\_ConversionMode:

Number Of Conversion **5 \***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 4

Sampling Time **480 Cycles \***

Rank **2 \***

Channel **Channel 7 \***

Sampling Time **480 Cycles \***

Rank **3 \***

Channel **Channel 14 \***

Sampling Time **480 Cycles \***

Rank **4 \***

Channel **Channel Temperature Sensor \***

Sampling Time **480 Cycles \***

Rank **5 \***

|                                     |                          |
|-------------------------------------|--------------------------|
| Channel                             | <b>Channel Vrefint *</b> |
| Sampling Time                       | <b>480 Cycles *</b>      |
| <b>ADC_Injected_ConversionMode:</b> |                          |
| Number Of Conversions               | 0                        |
| <b>WatchDog:</b>                    |                          |
| Enable Analog WatchDog Mode         | false                    |

### 3.2. CAN1

#### mode: Activated

##### 3.2.1. Parameter Settings:

###### Bit Timings Parameters:

|                              |                            |
|------------------------------|----------------------------|
| Prescaler (for Time Quantum) | <b>12 *</b>                |
| Time Quantum                 | <b>285.7142857142857 *</b> |
| Time Quanta in Bit Segment 1 | <b>5 Times *</b>           |
| Time Quanta in Bit Segment 2 | 1 Time                     |
| Time for one Bit             | <b>2000 *</b>              |
| Baud Rate                    | <b>500000 *</b>            |
| ReSynchronization Jump Width | 1 Time                     |

###### Basic Parameters:

|                                   |         |
|-----------------------------------|---------|
| Time Triggered Communication Mode | Disable |
| Automatic Bus-Off Management      | Disable |
| Automatic Wake-Up Mode            | Disable |
| Automatic Retransmission          | Disable |
| Receive Fifo Locked Mode          | Disable |
| Transmit Fifo Priority            | Disable |

###### Advanced Parameters:

|                |        |
|----------------|--------|
| Operating Mode | Normal |
|----------------|--------|

### 3.3. RCC

#### High Speed Clock (HSE): Crystal/Ceramic Resonator

##### 3.3.1. Parameter Settings:

###### System Parameters:

|                   |         |
|-------------------|---------|
| VDD voltage (V)   | 3.3     |
| Instruction Cache | Enabled |

|                   |                    |
|-------------------|--------------------|
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 5 WS (6 CPU cycle) |

**RCC Parameters:**

|                                |      |
|--------------------------------|------|
| HSI Calibration Value          | 16   |
| HSE Startup Timeout Value (ms) | 100  |
| LSE Startup Timeout Value (ms) | 5000 |

**Power Parameters:**

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

### 3.4. SYS

**Debug: Serial Wire**

**Timebase Source: TIM12**

### 3.5. TIM2

**Channel1: Output Compare No Output**

**Channel2: Input Capture direct mode**

**Channel3: Input Capture direct mode**

**Channel4: Input Capture direct mode**

**3.5.1. Parameter Settings:**

**Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Counter Period (AutoReload Register - 32 bits value ) | 4294967295  |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

**Trigger Output (TRGO) Parameters:**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

**Output Compare No Output Channel 1:**

|                        |                               |
|------------------------|-------------------------------|
| Mode                   | Frozen (used for Timing base) |
| Pulse (32 bits value)  | 0                             |
| Output compare preload | Disable                       |
| CH Polarity            | High                          |

**Input Capture Channel 2:**

|                    |                     |
|--------------------|---------------------|
| Polarity Selection | <b>Both Edges *</b> |
| IC Selection       | Direct              |

|                             |             |
|-----------------------------|-------------|
| Prescaler Division Ratio    | No division |
| Input Filter (4 bits value) | 0           |

#### Input Capture Channel 3:

|                             |                     |
|-----------------------------|---------------------|
| Polarity Selection          | <b>Both Edges *</b> |
| IC Selection                | Direct              |
| Prescaler Division Ratio    | No division         |
| Input Filter (4 bits value) | 0                   |

#### Input Capture Channel 4:

|                             |                     |
|-----------------------------|---------------------|
| Polarity Selection          | <b>Both Edges *</b> |
| IC Selection                | Direct              |
| Prescaler Division Ratio    | No division         |
| Input Filter (4 bits value) | 0                   |

### 3.6. TIM3

**Clock Source : Internal Clock**

**Channel1: Input Capture direct mode**

**Channel2: Input Capture direct mode**

**Channel3: Input Capture direct mode**

**Channel4: Input Capture direct mode**

#### 3.6.1. Parameter Settings:

##### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Counter Period (AutoReload Register - 16 bits value ) | 65535       |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

##### Trigger Output (TRGO) Parameters:

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

#### Input Capture Channel 1:

|                             |                       |
|-----------------------------|-----------------------|
| Polarity Selection          | <b>Falling Edge *</b> |
| IC Selection                | Direct                |
| Prescaler Division Ratio    | No division           |
| Input Filter (4 bits value) | <b>15 *</b>           |

#### Input Capture Channel 2:

|                    |                       |
|--------------------|-----------------------|
| Polarity Selection | <b>Falling Edge *</b> |
| IC Selection       | Direct                |

|                             |             |
|-----------------------------|-------------|
| Prescaler Division Ratio    | No division |
| Input Filter (4 bits value) | <b>15 *</b> |

#### Input Capture Channel 3:

|                             |                       |
|-----------------------------|-----------------------|
| Polarity Selection          | <b>Falling Edge *</b> |
| IC Selection                | Direct                |
| Prescaler Division Ratio    | No division           |
| Input Filter (4 bits value) | <b>15 *</b>           |

#### Input Capture Channel 4:

|                             |                       |
|-----------------------------|-----------------------|
| Polarity Selection          | <b>Falling Edge *</b> |
| IC Selection                | Direct                |
| Prescaler Division Ratio    | No division           |
| Input Filter (4 bits value) | <b>15 *</b>           |

### 3.7. TIM4

**Clock Source : Internal Clock**

**Channel1: Output Compare No Output**

#### 3.7.1. Parameter Settings:

##### Counter Settings:

|                                                       |                |
|-------------------------------------------------------|----------------|
| Prescaler (PSC - 16 bits value)                       | <b>13125 *</b> |
| Counter Mode                                          | Up             |
| Counter Period (AutoReload Register - 16 bits value ) | 65535          |
| Internal Clock Division (CKD)                         | No Division    |
| auto-reload preload                                   | Disable        |

##### Trigger Output (TRGO) Parameters:

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

##### Output Compare No Output Channel 1:

|                        |                               |
|------------------------|-------------------------------|
| Mode                   | Frozen (used for Timing base) |
| Pulse (16 bits value)  | 0                             |
| Output compare preload | Disable                       |
| CH Polarity            | High                          |

### 3.8. TIM5

**Combined Channels: Encoder Mode**



### 3.8.1. Parameter Settings:

#### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Counter Period (AutoReload Register - 32 bits value ) | 4294967295  |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

#### Trigger Output (TRGO) Parameters:

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

#### Encoder:

|                                    |                  |
|------------------------------------|------------------|
| Encoder Mode                       | Encoder Mode TI1 |
| ____ Parameters for Channel 1 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |
| ____ Parameters for Channel 2 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |

## 3.9. TIM9

### mode: Clock Source

### Channel1: PWM Generation CH1

### mode: One Pulse Mode

### 3.9.1. Parameter Settings:

#### Counter Settings:

|                                                       |               |
|-------------------------------------------------------|---------------|
| Prescaler (PSC - 16 bits value)                       | 0             |
| Counter Mode                                          | Up            |
| Counter Period (AutoReload Register - 16 bits value ) | <b>1680 *</b> |
| Internal Clock Division (CKD)                         | No Division   |
| auto-reload preload                                   | Disable       |

#### PWM Generation Channel 1:

|                       |              |
|-----------------------|--------------|
| Mode                  | PWM mode 1   |
| Pulse (16 bits value) | <b>504 *</b> |

|                        |         |
|------------------------|---------|
| Output compare preload | Enable  |
| Fast Mode              | Disable |
| CH Polarity            | High    |

### 3.10. TIM13

**mode: Activated**

**Channel1: PWM Generation CH1**

#### 3.10.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |                |
|-------------------------------------------------------|----------------|
| Prescaler (PSC - 16 bits value)                       | 0              |
| Counter Mode                                          | Up             |
| Counter Period (AutoReload Register - 16 bits value ) | <b>21000 *</b> |
| Internal Clock Division (CKD)                         | No Division    |
| auto-reload preload                                   | Disable        |

##### **PWM Generation Channel 1:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

### 3.11. USART3

**Mode: Asynchronous**

#### 3.11.1. Parameter Settings:

##### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

##### **Advanced Parameters:**

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |

### 3.12. FREERTOS

#### Interface: CMSIS\_V1

##### 3.12.1. Config parameters:

###### API:

FreeRTOS API CMSIS v1

###### Versions:

FreeRTOS version 10.3.1

CMSIS-RTOS version 1.02

###### MPU/FPU:

ENABLE\_MPU Disabled

ENABLE\_FPU Disabled

###### Kernel settings:

USE\_PREEMPTION Enabled

CPU\_CLOCK\_HZ SystemCoreClock

TICK\_RATE\_HZ **512 \***

MAX\_PRIORITIES 7

MINIMAL\_STACK\_SIZE **96 \***

MAX\_TASK\_NAME\_LEN 16

USE\_16\_BIT\_TICKS Disabled

IDLE\_SHOULD\_YIELD Enabled

USE\_MUTEXES Enabled

USE\_RECURSIVE\_MUTEXES Disabled

USE\_COUNTING\_SEMAPHORES Disabled

QUEUE\_REGISTRY\_SIZE **12 \***

USE\_APPLICATION\_TASK\_TAG Disabled

ENABLE\_BACKWARD\_COMPATIBILITY Enabled

USE\_PORT\_OPTIMISED\_TASK\_SELECTION Enabled

USE\_TICKLESS\_IDLE Disabled

USE\_TASK\_NOTIFICATIONS Enabled

RECORD\_STACK\_HIGH\_ADDRESS Disabled

###### Memory management settings:

Memory Allocation Dynamic / Static

TOTAL\_HEAP\_SIZE **32768 \***

Memory Management scheme heap\_4

###### Hook function related definitions:

USE\_IDLE\_HOOK Disabled

USE\_TICK\_HOOK Disabled

USE\_MALLOC\_FAILED\_HOOK Disabled

USE\_DAEMON\_TASK\_STARTUP\_HOOK Disabled

CHECK\_FOR\_STACK\_OVERFLOW

**Option2 \***

**Run time and task stats gathering related definitions:**

GENERATE\_RUN\_TIME\_STATS Disabled

USE\_TRACE\_FACILITY Disabled

USE\_STATS\_FORMATTING\_FUNCTIONS Disabled

**Co-routine related definitions:**

USE\_CO\_ROUTINES Disabled

MAX\_CO\_ROUTINE\_PRIORITIES 2

**Software timer definitions:**

USE\_TIMERS **Enabled \***

TIMER\_TASK\_PRIORITY 2

TIMER\_QUEUE\_LENGTH 10

TIMER\_TASK\_STACK\_DEPTH 192

**Interrupt nesting behaviour configuration:**

LIBRARY\_LOWEST\_INTERRUPT\_PRIORITY 15

LIBRARY\_MAX\_SYSCALL\_INTERRUPT\_PRIORITY 5

**Added with 10.2.1 support:**

MESSAGE\_BUFFER\_LENGTH\_TYPE size\_t

USE\_POSIX\_ERRNO Disabled

3.12.2. Include parameters:

**Include definitions:**

vTaskPrioritySet Enabled

uxTaskPriorityGet Enabled

vTaskDelete **Disabled \***

vTaskCleanUpResources Disabled

vTaskSuspend Enabled

vTaskDelayUntil **Enabled \***

vTaskDelay Enabled

xTaskGetSchedulerState Enabled

xTaskResumeFromISR Enabled

xQueueGetMutexHolder Disabled

xSemaphoreGetMutexHolder Disabled

pcTaskGetTaskName Disabled

uxTaskGetStackHighWaterMark **Enabled \***

xTaskGetCurrentTaskHandle **Enabled \***

eTaskGetState Disabled

xEventGroupSetBitFromISR Disabled

xTimerPendFunctionCall Disabled

|                              |          |
|------------------------------|----------|
| xTaskAbortDelay              | Disabled |
| xTaskGetHandle               | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.12.3. Advanced settings:

#### **Newlib settings (see parameter description first):**

|                      |          |
|----------------------|----------|
| USE_NEWLIB_REENTRANT | Disabled |
|----------------------|----------|

#### **Project settings (see parameter description first):**

|                       |         |
|-----------------------|---------|
| Use FW pack heap file | Enabled |
|-----------------------|---------|

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP     | Pin         | Signal         | GPIO mode                       | GPIO pull/up pull down      | Max Speed   | User Label     |
|--------|-------------|----------------|---------------------------------|-----------------------------|-------------|----------------|
| ADC1   | PA4         | ADC1_IN4       | Analog mode                     | No pull-up and no pull-down | n/a         | stepperV       |
|        | PA7         | ADC1_IN7       | Analog mode                     | No pull-up and no pull-down | n/a         | Reg5v          |
|        | PC4         | ADC1_IN14      | Analog mode                     | No pull-up and no pull-down | n/a         | SpareV         |
| CAN1   | PD0         | CAN1_RX        | Alternate Function Push Pull    | No pull-up and no pull-down | Very High * |                |
|        | PD1         | CAN1_TX        | Alternate Function Push Pull    | No pull-up and no pull-down | Very High * |                |
| RCC    | PH0-OSC_IN  | RCC_OSC_IN     | n/a                             | n/a                         | n/a         |                |
|        | PH1-OSC_OUT | RCC_OSC_OUT    | n/a                             | n/a                         | n/a         |                |
| SYS    | PA13        | SYS_JTMS-SWDIO | n/a                             | n/a                         | n/a         |                |
|        | PA14        | SYS_JTCK-SWCLK | n/a                             | n/a                         | n/a         |                |
| TIM2   | PA2         | TIM2_CH3       | Alternate Function Open Drain * | No pull-up and no pull-down | Low         | encodertimeA   |
|        | PA3         | TIM2_CH4       | Alternate Function Open Drain * | No pull-up and no pull-down | Low         | encodertimeB   |
|        | PB3         | TIM2_CH2       | Alternate Function Open Drain * | No pull-up and no pull-down | Low         | encodertimeZ   |
| TIM3   | PC6         | TIM3_CH1       | Alternate Function Push Pull    | Pull-up *                   | Low         | LimitSw_MSN_NO |
|        | PC7         | TIM3_CH2       | Alternate Function Push Pull    | Pull-up *                   | Low         | LimitSw_MSN_NC |
|        | PC8         | TIM3_CH3       | Alternate Function Push Pull    | Pull-up *                   | Low         | LimitSw_MS_NO  |
|        | PC9         | TIM3_CH4       | Alternate Function Push Pull    | Pull-up *                   | Low         | LimitSw_MS_NC  |
| TIM5   | PA0-WKUP    | TIM5_CH1       | Alternate Function Push Pull    | No pull-up and no pull-down | Low         | encodetrA      |
|        | PA1         | TIM5_CH2       | Alternate Function Push Pull    | No pull-up and no pull-down | Low         | encodetrB      |
| TIM9   | PE5         | TIM9_CH1       | Alternate Function Push Pull    | Pull-up *                   | Very High * | Stepper_PU     |
| TIM13  | PA6         | TIM13_CH1      | Alternate Function Push Pull    | No pull-up and no pull-down | Very High * | fieldfetdrive  |
| USART3 | PC10        | USART3_TX      | Alternate Function Push Pull    | No pull-up and no pull-down | Very High * |                |
|        | PC11        | USART3_RX      | Alternate Function Push Pull    | No pull-up and no pull-down | Very High   |                |

| IP   | Pin  | Signal      | GPIO mode                                                          | GPIO pull/up pull down      | Max Speed       | User Label      |
|------|------|-------------|--------------------------------------------------------------------|-----------------------------|-----------------|-----------------|
|      |      |             |                                                                    |                             | *               |                 |
| GPIO | PB0  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | <b>Medium *</b> | Stepper_DR      |
|      | PB1  | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low             | Stepper_MF      |
|      | PE7  | GPIO_Input  | Input mode                                                         | <b>Pull-up *</b>            | n/a             | ManualSw_MSN_NO |
|      | PE8  | GPIO_Input  | Input mode                                                         | <b>Pull-up *</b>            | n/a             | ManualSw_MS_NO  |
|      | PE9  | GPIO_Input  | Input mode                                                         | <b>Pull-up *</b>            | n/a             | ManualSw_NO     |
|      | PE10 | GPIO_EXTI10 | <b>External Interrupt Mode with Falling edge trigger detection</b> | <b>Pull-up *</b>            | n/a             | LimitSw_MSN_NO  |
|      | PE11 | GPIO_EXTI11 | <b>External Interrupt Mode with Falling edge trigger detection</b> | <b>Pull-up *</b>            | n/a             | LimitSw_MSN_NC  |
|      | PE12 | GPIO_EXTI12 | <b>External Interrupt Mode with Falling edge trigger detection</b> | <b>Pull-up *</b>            | n/a             | LimitSw_MS_NO   |
|      | PE13 | GPIO_EXTI13 | <b>External Interrupt Mode with Falling edge trigger detection</b> | <b>Pull-up *</b>            | n/a             | LimitSw_MS_NC   |
|      | PE14 | GPIO_Input  | Input mode                                                         | <b>Pull-up *</b>            | n/a             | OverrunSwes_NO  |
|      | PD12 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low             | LED GREEN       |
|      | PD13 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low             | LED ORANGE      |
|      | PD14 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low             | LED RED         |
|      | PD15 | GPIO_Output | Output Push Pull                                                   | No pull-up and no pull-down | Low             | LED BLUE        |

## 4.2. DMA configuration

| DMA request | Stream       | Direction            | Priority |
|-------------|--------------|----------------------|----------|
| USART3_RX   | DMA1_Stream1 | Peripheral To Memory | Low      |
| USART3_TX   | DMA1_Stream3 | Memory To Peripheral | Low      |
| ADC1        | DMA2_Stream0 | Peripheral To Memory | Low      |

### USART3\_RX: DMA1\_Stream1 DMA request Settings:

Mode: Normal  
 Use fifo: Disable  
 Peripheral Increment: Disable  
 Memory Increment: **Enable \***  
 Peripheral Data Width: Byte  
 Memory Data Width: Byte

### USART3\_TX: DMA1\_Stream3 DMA request Settings:

Mode: Normal  
 Use fifo: Disable  
 Peripheral Increment: Disable  
 Memory Increment: **Enable \***  
 Peripheral Data Width: Byte  
 Memory Data Width: Byte

### ADC1: DMA2\_Stream0 DMA request Settings:

Mode: **Circular \***  
 Use fifo: Disable  
 Peripheral Increment: Disable  
 Memory Increment: **Enable \***  
 Peripheral Data Width: Half Word  
 Memory Data Width: Half Word



### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                  | Enable | Preenmption Priority | SubPriority |
|--------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                           | true   | 0                    | 0           |
| Hard fault interrupt                             | true   | 0                    | 0           |
| Memory management fault                          | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault             | true   | 0                    | 0           |
| Undefined instruction or illegal state           | true   | 0                    | 0           |
| System service call via SWI instruction          | true   | 0                    | 0           |
| Debug monitor                                    | true   | 0                    | 0           |
| Pendable request for system service              | true   | 15                   | 0           |
| System tick timer                                | true   | 15                   | 0           |
| DMA1 stream1 global interrupt                    | true   | 10                   | 0           |
| DMA1 stream3 global interrupt                    | true   | 10                   | 0           |
| ADC1, ADC2 and ADC3 global interrupts            | true   | 7                    | 0           |
| CAN1 TX interrupts                               | true   | 7                    | 0           |
| CAN1 RX0 interrupts                              | true   | 7                    | 0           |
| CAN1 RX1 interrupt                               | true   | 7                    | 0           |
| TIM1 break interrupt and TIM9 global interrupt   | true   | 6                    | 0           |
| TIM2 global interrupt                            | true   | 2                    | 0           |
| TIM4 global interrupt                            | true   | 2                    | 0           |
| USART3 global interrupt                          | true   | 8                    | 0           |
| EXTI line[15:10] interrupts                      | true   | 5                    | 0           |
| TIM8 break interrupt and TIM12 global interrupt  | true   | 0                    | 0           |
| DMA2 stream0 global interrupt                    | true   | 7                    | 0           |
| PVD interrupt through EXTI line 16               | unused |                      |             |
| Flash global interrupt                           | unused |                      |             |
| RCC global interrupt                             | unused |                      |             |
| CAN1 SCE interrupt                               | unused |                      |             |
| TIM3 global interrupt                            | unused |                      |             |
| TIM8 update interrupt and TIM13 global interrupt | unused |                      |             |
| TIM5 global interrupt                            | unused |                      |             |
| FPU global interrupt                             | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt  | false                             | true                 | false            |
| Hard fault interrupt    | false                             | true                 | false            |
|                         |                                   |                      |                  |

| Enabled interrupt Table                         | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-------------------------------------------------|-----------------------------------|----------------------|------------------|
| Memory management fault                         | false                             | true                 | false            |
| Pre-fetch fault, memory access fault            | false                             | true                 | false            |
| Undefined instruction or illegal state          | false                             | true                 | false            |
| System service call via SWI instruction         | false                             | false                | false            |
| Debug monitor                                   | false                             | true                 | false            |
| Pendable request for system service             | false                             | false                | false            |
| System tick timer                               | false                             | false                | true             |
| DMA1 stream1 global interrupt                   | false                             | true                 | true             |
| DMA1 stream3 global interrupt                   | false                             | true                 | true             |
| ADC1, ADC2 and ADC3 global interrupts           | false                             | true                 | true             |
| CAN1 TX interrupts                              | false                             | true                 | true             |
| CAN1 RX0 interrupts                             | false                             | true                 | true             |
| CAN1 RX1 interrupt                              | false                             | true                 | true             |
| TIM1 break interrupt and TIM9 global interrupt  | false                             | true                 | true             |
| TIM2 global interrupt                           | false                             | true                 | true             |
| TIM4 global interrupt                           | false                             | true                 | true             |
| USART3 global interrupt                         | false                             | true                 | true             |
| EXTI line[15:10] interrupts                     | false                             | true                 | true             |
| TIM8 break interrupt and TIM12 global interrupt | false                             | true                 | true             |
| DMA2 stream0 global interrupt                   | false                             | true                 | true             |

**\* User modified value**

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

| Middleware  |        |         |              |            |          |           |
|-------------|--------|---------|--------------|------------|----------|-----------|
| FREERTOS ✓  |        |         |              |            |          |           |
| System Core | Analog | Timers  | Connectivity | Multimedia | Security | Computing |
| DMA ✓       | ADC1 ✓ | TIM2 ✓  | CAN1 ✓       |            |          |           |
| GPIO ✓      |        | TIM3 ✓  | USART3 ✓     |            |          |           |
| NVIC ✓      |        | TIM4 ✓  |              |            |          |           |
| RCC ✓       |        | TIM5 ✓  |              |            |          |           |
| SYS ✓       |        | TIM9 ✓  |              |            |          |           |
|             |        | TIM13 ✓ |              |            |          |           |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32f405-415_407-417_bsdl.zip">https://www.st.com/resource/en/bsdl_model/stm32f405-415_407-417_bsdl.zip</a>                                                                                                           |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32f405-415_407-417_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32f405-415_407-417_ibis.zip</a>                                                                                                           |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32f4-svd.zip">https://www.st.com/resource/en/svd/stm32f4-svd.zip</a>                                                                                                                                                       |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                       |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf</a>                                                         |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                       |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                               |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstmcsuite.pdf">https://www.st.com/resource/en/flyer/flstmcsuite.pdf</a>                                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                 |
| Product Certifications  | <a href="https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf">https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf</a>                                                                                       |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a> |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf</a>     |

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| Programming<br>Manuals        | <a href="https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                                                 |
| Reference<br>Manuals          | <a href="https://www.st.com/resource/en/reference_manual/rm0090-stm32f405415-stm32f407417-stm32f427437-and-stm32f429439-advanced-armbased-32bit-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/reference_manual/rm0090-stm32f405415-stm32f407417-stm32f427437-and-stm32f429439-advanced-armbased-32bit-mcus-stmicroelectronics.pdf</a>           |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn0516-overview-of-the-stm32f0xf100xxf103xx-and-stm32f2xxf30xf4xx-mcus-pmsm-singledual-foc-sdk-v40-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn0516-overview-of-the-stm32f0xf100xxf103xx-and-stm32f2xxf30xf4xx-mcus-pmsm-singledual-foc-sdk-v40-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf</a>                                     |
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| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf</a>                   |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-</a>                                                                   |

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